ASHWIN SHARMA

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DATA & BACKEND ENGINEERING | PYTHON • SQL • POSTGRESQL • DATA PIPELINES

PROFESSIONAL SUMMARY

Software and AI/ML Engineer with 2+ years of combined industry experience, including an engineering internship and full-time work, building Python backends, PostgreSQL-backed applications, API integrations and data-processing workflows. Experienced in relational data modelling, schema migrations, document validation, dataset integration, data-driven analytics and cloud deployment. Skilled in Python, SQL, pandas, PostgreSQL, FastAPI, Django and GCP, with a focus on reliable data pipelines, data quality and scalable enterprise solutions.

TECHNICAL SKILLS

Programming & data: Python, SQL, pandas, NumPy

Databases: PostgreSQL, MySQL, SQLAlchemy, Alembic schema migrations

Backend & APIs: FastAPI, Django, Django REST Framework, asynchronous backend development

Data workflows: Data processing, dataset integration, annotation workflows, document validation, API-based integration

AI/ML & analytics: scikit-learn, TensorFlow, Hugging Face Transformers, LangChain, Google Vertex AI/Gemini, data-driven dashboards

Web & deployment: TypeScript, React, Vite, Git, GitHub, GCP, Vercel, Render, Neon, Gunicorn

Generative AI & Machine Learning: Generative AI, LLMs, Retrieval-Augmented Generation (RAG), Prompt Engineering, AI API Integration

AI-Assisted Development: AI Coding Assistants, AI-Assisted Debugging, Code Generation, Development Workflow Automation

EXPERIENCE

Software Engineering Intern | CARD (Center for Astrology Research & Development)

Sep 2024 – Mar 2025

Selected System: AstroLedger – Vedic Astrology Practice Management Suite

Technologies: Python, Django, Django Channels/Daphne, SQLite, React, TypeScript, IndexedDB, Electron, Capacitor

• Developed an offline-first, cross-platform practice management application for consultation records, client details, horoscopic information and notes, with local-first data storage.

• Implemented a desktop-led local sync architecture using REST APIs and WebSockets, with QR-based device pairing over private Wi-Fi and an offline mutation queue for mobile clients.

• Designed workflows for consultation history, structured observations and remedies, bilingual Vedic reference data, and formatted A4 consultation summaries.

• Packaged the application for Windows and Android, including a standalone backend distribution and Android APK build workflow.

Associate AI/ML Engineer | Gaude Business & Infrastructure Solutions

May 2025 – Present

- Develop Python-based workflows for AI-enabled content generation, document processing, validation and automated output production.
- Implemented Retrieval-Augmented Generation (RAG) workflows with LangChain, focusing on contextual retrieval and structured outputs.
- Built end-to-end pipelines that connect text generation, illustration generation and PDF export, integrating external AI services through APIs.
- Improved consistency and efficiency by refining prompts, orchestration, context handling, caching and API-call patterns.
- Contributed to an internal LLM-powered content-generation and workflow-orchestration system.

Selected System: MUDU NeoBench – Corporate Learning Platform

- Built a multi-organisation learning platform covering courses, live sessions, learner progress, goals, assessments, notifications and analytics.
- Designed FastAPI backend services with PostgreSQL, async SQLAlchemy, JWT authentication, role/organisation-scoped access controls and Alembic-managed schema migrations.
- Implemented course, quiz, assignment and goal-tracking workflows, including grading rules, submission validation and event-driven notifications.

- Integrated scheduling and AI-assisted content/quiz generation through external services; kept dashboard and analytics content data-driven for routine rendering.

Selected System: Mythora Ecosystem – AI Publishing & Commerce Platform

- Developed connected Django/React applications for AI-assisted publishing and a cultural streaming and e-commerce storefront.
- Built workflows for story and activity-book generation, illustration, audiobook production and print-ready PDF fulfilment using third-party AI APIs.
- Optimised service usage through prompt tuning, caching and routing image-generation requests according to cost/quality trade-offs.
- Implemented API-based integration between applications and role-based workflows for publishing, fulfilment and commerce.

SELECTED PROJECTS

AuraFit – AI Pantry & Fitness Tracker | *React, Django REST Framework, PostgreSQL, Gemini AI*

- Built a responsive PWA with pantry management, meal planning and fitness tracking, using a React frontend and Django REST backend.
- Integrated Gemini AI for meal-plan generation based on pantry contents and nutritional targets; implemented offline app-shell caching and service-worker support.

Carnatic Raga Identification & Kriti Segmentation Annotation | *Python, model fine-tuning*

- Designed an annotation tool to segment Carnatic kritis into structural sections and label each segment with its raga, including raga transitions.
- Built a data pipeline combining user-annotated segments, uploaded pre-annotated datasets and external sources to support unified model training; model fine-tuning and transition detection are in progress.

Detection of Smart Grey Hole Attack in VANET | *Python*

- Implemented a hybrid detection approach combining speed-based clustering and threshold-based classification, with central monitoring to validate local alerts and reduce false positives in simulations.

Gesture-Controlled Media | *Python, OpenCV, MediaPipe*

- Built a real-time hand-gesture recognition system for media playback controls; reported 90%+ recognition accuracy in testing.

EDUCATION

Bachelor of Technology (B.Tech), Computer Science & Engineering | Amrita Vishwa Vidyapeetham, Coimbatore | 2025 | CGPA: 6.61/10

Senior Secondary Education (Class XII) | Bhavan's Vidya Mandir, Ernakulam | 2021 | CGPA: 9.34

Secondary Education (Class X) | Bhavan's Vidya Mandir, Ernakulam | 2019 | CGPA: 9.24

Relevant coursework: Machine Learning, Data Mining, Computer Networks